

JAYASANKAR KUMAR SANTHIRANI

Thiruvananthapuram, Kerala, India

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Education

Vellore Institute of Technology - Chennai, India

Sept. 2021 – May 2025

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.66 / 10.0

- **Bachelor Thesis:** *Luxe Concierge: An LLM-Powered Architecture For Attribute-Aware Conversational Recommendation With Contextual Reranking*
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Database Management Systems, Human-Computer Interaction, Computer Networks, Artificial Intelligence, Web Programming, Theory of Computation

Experience

Noctis Tide Solutions

October 2024 – December 2024

UI/UX Intern

Chennai, India

- Contributed to the design and implementation of user-focused web application interfaces and game assets to enhance digital experiences, contributing to a **3% increase in early user engagement metrics**.
- Developed interactive and intuitive web UI/UX prototypes, improving application navigation flow and user retention by **4% through usability testing and feedback integration**.
- Implemented design optimizations for visual consistency and adherence to UI/UX best practices across web applications, **enhancing team productivity by 5%**.
- Collaborated with cross-functional teams to refine requirements and ensure effective user interface solutions.

Keltron KSG

Sept. 2023 – Oct. 2023

Networking & Cybersecurity Intern

Trivandrum, Kerala

- Configured, monitored, and maintained secure network systems using tools like Cisco Packet Tracer, Wireshark, and SolarWinds, contributing to the development of secure code practices.
- Applied frameworks such as NIST Cybersecurity Framework and tools like nmap, sqlmap, and Snort to identify vulnerabilities, deploy security protocols, and ensure data protection, informing secure system development.

Trizent Technologies

July 2022 – August 2022

Software Testing Intern

Trivandrum, Kerala

- Automated web application testing processes by developing and executing **25+ automated test cases** using **Robot Framework**, significantly contributing to continuous improvement initiatives.
- Collaborated effectively with cross-functional development teams to identify, document, and resolve **10+ UI and functional bugs**, ensuring system stability and seamless development workflow.
- Conducted systematic regression testing across **3 major dashboard updates**, reducing recurring issues and enhancing overall platform reliability.

Projects

School Website – Informational Site for Educational Institution (*React.js, Tailwind CSS, Express.js, Supabase*):

Developed and deployed a responsive website to showcase school information and announcements. Built with React and Tailwind CSS for frontend, Express.js for backend routing, and Supabase for serverless data management. **[Live Website]**

LuxeConcierge - Conversational Recommender System (*Python, DeepSeek, NVIDIA NIM*): Architected and developed a full-stack, LLM-powered conversational recommender system in Python, featuring complex API integrations for enhanced contextual understanding and scalable solutions. **[GitHub]**

CerviLens – AI-Powered Cervical Cancer Screening Interface (*HTML, CSS, JavaScript, WebRTC, React.js*):

Engineered a responsive frontend interface for a real-time cervical cancer screening tool. Implemented camera integration using WebRTC to capture and display procedural images, enabling seamless classification into Syn-1, Syn-2, Syn-3, positive, or negative. Designed dynamic components to visualize automated AI-powered diagnosis and generated structured screening reports to enhance clinical decision-making. **[Project]**

ItinerEase - AI Travel Planner (*HTML, CSS, JavaScript, Figma*): Developed a responsive frontend for an AI-powered travel planner integrating a chatbot-driven conversational interface for itinerary generation. Designed intuitive user interfaces aligned with modern UX principles to enhance engagement and trip customization. Conducted usability testing to optimize interaction flow, improving user satisfaction metrics by 12%. **[GitHub]**

Fetal Health Classification System (*Python*): Built a machine learning ensemble model to classify fetal health status into normal, suspect, or pathological categories. Incorporated continual learning algorithms and explainability features to enhance adaptability and transparency. Achieved a validation accuracy of over 94% using real-world cardiotocographic datasets. [\[GitHub\]](#)

Network Intrusion Detection System (*Python*): Designed and implemented a high-accuracy intrusion detection system using XGBoost with optimized feature selection techniques. Addressed challenges related to data imbalance and attack diversity, achieving 98% classification accuracy on the NSL-KDD dataset. Conducted extensive model validation to minimize false positives and ensure system robustness.

Technical Skills

- Programming Languages:** Python, JavaScript, HTML, CSS, SQL, Java, C, C++
Web Technologies & Frameworks: React (MERN Stack), RESTful APIs, GraphQL, Node.js, Flask, Django (conceptual), Figma, Framer, UI/UX Design, Responsive Web Design
Databases: Relational Databases, SQL, Basic Data Modeling
Cloud Platforms: Google Cloud Platform (GCP), Firebase, Exposure to AWS concepts
Version Control & Methodologies: Git, Agile, Scrum
Automation Tools: Google Apps Script, Robot Framework, Python Scripting, Webhooks
Security & Networking: NIST Cybersecurity Framework, Wireshark, nmap, sqlmap, Snort, Splunk, OpenVAS, Burp Suite, Cisco Packet Tracer
Developer Tools: VS Code, Eclipse

Leadership / Extracurricular

Photography Club

Design Head

- Directed all design initiatives for the college photography club, ensuring cohesive visual representation across events and projects.
- Designed promotional materials, including posters, banners, and digital content, to showcase activities and attract engagement.
- Collaborated with photographers and organizers to develop event-specific branding, enhancing overall aesthetic appeal and professional presentation.

April 2023 – April 2024

VIT Chennai

Achievements / Scholarships

- Scholarship Programme for Diaspora Children awarded to 150 meritorious students worldwide.
- First Runners Up in International UI/UX Competition organized by Universiti Putra Malaysia and Carlo Rino.